

- Description

This module manages an energy consumption monitoring system for factories, ensuring that:

1. **Energy usage is recorded periodically and compared against predefined limits.**
2. **If energy consumption exceeds the limit, an overuse warning is triggered.**
3. **The system recommends energy-saving measures when excessive consumption is detected.**
4. **A log of all energy usage data and alerts is maintained for further analysis.**

This system helps factories optimize energy consumption, reduce costs, and promote sustainability.

- Specification

module FactoryEnergyMonitoring

var: energyUsed, productionOutput, efficiency, maxAllowedUsage: real

var energyLog: sequence of string

process recordUsage(energyUsed: real, productionOutput: real)

functional scenario form:

energyUsed <= maxAllowedUsage && energyLog = energyLog $\oplus$ ("Usage Recorded: " + energyUsed + " kWh"), Energy usage recorded successfully.

||

energyUsed > maxAllowedUsage && energyLog = energyLog $\oplus$ ("Overuse Warning: " + energyUsed + " kWh"), Overuse warning triggered.

end process

end module

- Java Code

```
package code;
import java.util.*;
class EnergyMonitor {
    private double maxAllowedUsage;
    private List<String> energyLog;
    public EnergyMonitor(double maxAllowedUsage) {
        this.maxAllowedUsage = maxAllowedUsage;
        this.energyLog = new ArrayList<>();
    }
    public void recordUsage(double energyUsed, double productionOutput)
    {
        if (energyUsed <= 0 || productionOutput <= 0) {
            System.out.println("Invalid input: Energy usage and product
ion output must be greater than zero.");
            return;
        }
    }
}
```

```

        double efficiency = productionOutput / energyUsed;
        String message = String.format("Energy Used: %.1f kWh | Production Output: %.1f units | Efficiency: %.2f units/kWh",
                                         energyUsed, productionOutput, efficiency);
        if (energyUsed > maxAllowedUsage) {
            message += " | WARNING: Energy Overuse Detected! Consider efficiency improvements.";
        }
        System.out.println(message);
        energyLog.add(message);
    }

    public void printEnergyLog() {
        System.out.println("\nEnergy Consumption Log:");
        for (String log : energyLog) {
            System.out.println(log);
        }
    }
}

public class c26_FactoryEnergySystem {
    public static void main(String[] args) {
        EnergyMonitor monitor = new EnergyMonitor(1000.0);
        // testcase-VT:
        // monitor.recordUsage(-929.93, 3398.41);
        // monitor.recordUsage(4728.84, 1585.89);
        // monitor.recordUsage(3544.78, 437.15);
        // monitor.recordUsage(-775.56, 4326.12);
        // monitor.recordUsage(3301.20, -4436.36);
        // monitor.recordUsage(-2355.64, -577.59);
        // monitor.recordUsage(2069.94, -4698.54);
        // monitor.recordUsage(-2554.84, -3989.47);
        // monitor.recordUsage(1411.68, -2635.97);
        // monitor.recordUsage(-1129.01, -4662.37);
        // monitor.recordUsage(-705.61, -2685.10);
        // monitor.recordUsage(-3679.25, -187.01);
        // monitor.recordUsage(2869.89, -3314.98);
        // monitor.recordUsage(19.73, -2089.05);
        // monitor.recordUsage(1235.30, -4042.63);
        // monitor.recordUsage(-1652.31, -3768.61);
        // monitor.recordUsage(-881.17, -2702.65);
        // monitor.recordUsage(-230.48, -1781.02);
        // monitor.recordUsage(4113.51, -3454.50);
        // monitor.recordUsage(1239.88, 114.39);
        // monitor.recordUsage(-190.10, -4404.27);
    }
}

```

```
//      monitor.recordUsage(-4781.30,4745.49);
//      monitor.recordUsage(4871.58,766.79);
//      monitor.recordUsage(2752.15,-1090.90);
//      monitor.recordUsage(3914.09,765.53);
//      monitor.recordUsage(1184.62,1438.07);
//      monitor.recordUsage(-3366.22,2440.76);
//      monitor.recordUsage(3798.39,-4278.28);
//      monitor.recordUsage(-4679.81,1836.31);
//      monitor.recordUsage(4136.33,3960.34);
//      monitor.recordUsage(4828.81,1223.53);
//      monitor.recordUsage(4621.52,-4262.01);
//      monitor.recordUsage(-569.94,-4058.13);
//      monitor.recordUsage(4432.12,-622.88);
//      monitor.recordUsage(-3357.51,1324.03);
//      monitor.recordUsage(-1072.24,3660.60);
//      monitor.recordUsage(2835.98,3888.96);
//      monitor.recordUsage(409.32,-4106.45);
//      monitor.recordUsage(-2767.75,-2092.41);
//      monitor.recordUsage(-1276.05,4046.21);
//      monitor.recordUsage(-544.76,-923.40);
//      monitor.recordUsage(-366.00,-1048.28);
//      monitor.recordUsage(311.41,-2294.98);
//      monitor.recordUsage(1485.61,3736.57);
//      monitor.recordUsage(-498.23,-3251.17);
//      monitor.recordUsage(1857.71,4981.00);
//      monitor.recordUsage(210.36,-815.44);
//      monitor.recordUsage(-4495.43,-873.31);
//      monitor.recordUsage(-4896.96,1203.56);
//      monitor.recordUsage(1995.52,-3612.68);
//      monitor.recordUsage(-807.24,2171.48);
//      monitor.recordUsage(-3637.68,1917.63);
//      monitor.recordUsage(1668.89,27.42);
//      monitor.recordUsage(-853.00,-3740.07);
//      monitor.recordUsage(-565.00,-3434.03);
//      monitor.recordUsage(-236.86,3073.92);
//      monitor.recordUsage(4843.32,-2867.20);
//      monitor.recordUsage(-1657.20,-478.22);
//      monitor.recordUsage(-180.38,4299.27);
//      monitor.recordUsage(3022.87,-2257.78);
//      monitor.recordUsage(1037.47,4722.22);
//      monitor.recordUsage(-4761.60,1768.89);
//      monitor.recordUsage(-1600.59,-4053.44);
//      monitor.recordUsage(1520.17,-693.26);
//      monitor.recordUsage(-4841.63,-2334.68);
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]


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//      monitor.recordUsage(-3357.51,1324.03);
//      monitor.recordUsage(-1072.24,3660.60);
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//      monitor.recordUsage(409.32,-4106.45);
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//      monitor.recordUsage(-4761.60,1768.89);
//      monitor.recordUsage(-1600.59,-4053.44);
//      monitor.recordUsage(1520.17,-693.26);
//      monitor.recordUsage(-4841.63,-2334.68);
//      monitor.recordUsage(-2246.37,-2663.87);
//      monitor.recordUsage(4190.97,-1164.84);
//      monitor.recordUsage(-3840.32,-3770.34);
//      monitor.recordUsage(-855.00,-3733.84);
//      monitor.recordUsage(-656.29,-3267.95);
//      monitor.recordUsage(4850.67,-4241.27);
//      monitor.recordUsage(434.27,-3540.24);
//      monitor.recordUsage(-4496.82,-1401.37);
//      monitor.recordUsage(4642.46,-2261.43);
//      monitor.recordUsage(3182.98,4467.60);
//      monitor.recordUsage(-4446.87,3566.73);
//      monitor.recordUsage(2855.02,-4244.03);
//      monitor.recordUsage(3738.28,-1626.75);
```

```
//      monitor.recordUsage(-3643.85,2192.73);
//      monitor.recordUsage(1042.04,839.64);
//      monitor.recordUsage(-2411.31,-2656.35);
//      monitor.recordUsage(-798.40,-1700.44);
//      monitor.recordUsage(4466.82,-235.36);
//      monitor.recordUsage(-3776.09,4518.89);
//      monitor.recordUsage(-2793.75,2146.98);
//      monitor.recordUsage(2775.04,3665.36);
//      monitor.recordUsage(-2308.08,3525.30);
//      monitor.recordUsage(-583.75,-4119.68);
//      monitor.recordUsage(-3712.25,1898.60);
//      monitor.recordUsage(-2465.87,-1267.21);
//      monitor.recordUsage(-2337.14,-4422.94);
//      monitor.recordUsage(-4468.88,-4616.22);
//      monitor.recordUsage(-4653.17,4986.41);
//      monitor.recordUsage(1420.60,-3969.65);
//      monitor.recordUsage(3950.26,-236.26);
//      monitor.recordUsage(4424.71,-4843.33);
//      monitor.recordUsage(-791.76,-4290.14);
//      monitor.recordUsage(463.87,-2862.77);
//      monitor.recordUsage(-2781.18,3135.68);
//      monitor.recordUsage(-2126.09,-4837.62);
```

```
// testcase-UVT:
```

```
monitor.recordUsage(-929.93,3398.41);
monitor.recordUsage(4728.84,1585.89);
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monitor.recordUsage(4113.51,-3454.50);
monitor.recordUsage(1239.88,114.39);
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monitor.recordUsage(-190.10,-4404.27);
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monitor.recordUsage(4871.58,766.79);
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```

```

        monitor.recordUsage(-4841.63, -2334.68);
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        monitor.recordUsage(-3840.32, -3770.34);
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        monitor.recordUsage(-4496.82, -1401.37);
        monitor.recordUsage(4642.46, -2261.43);
        monitor.recordUsage(3182.98, 4467.60);
        monitor.recordUsage(-4446.87, 3566.73);
        monitor.recordUsage(2855.02, -4244.03);
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        monitor.recordUsage(-3776.09, 4518.89);
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        monitor.recordUsage(-2308.08, 3525.30);
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        monitor.recordUsage(-3712.25, 1898.60);
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        monitor.recordUsage(463.87, -2862.77);
        monitor.recordUsage(-2781.18, 3135.68);
        monitor.recordUsage(-2126.09, -4837.62);

//      monitor.printEnergyLog();
    }
}

```

- Coverage report

1. Test case number: 100
2. V-Method testing branch coverage: 87.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.8 %	7	835	842
✓ src	0.8 %	7	835	842
✓ code	0.8 %	7	835	842
✓ c26_FactoryEnergySystem.java	87.5 %	7	1	8
> EnergyMonitor	87.5 %	7	1	8
> c26_FactoryEnergySystem		0	0	0

3. Vibration testing instruction coverage: 99.4%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	2.8 %	491	17,128	17,619
✓ src	2.8 %	491	17,128	17,619
✓ code	2.8 %	491	17,128	17,619
✓ c26_FactoryEnergySystem.java	99.4 %	491	3	494
> c26_FactoryEnergySystem	99.3 %	408	3	411
> EnergyMonitor	100.0 %	83	0	83

4. Fuzz testing branch coverage: 62.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.6 %	5	837	842
✓ src	0.6 %	5	837	842
✓ code	0.6 %	5	837	842
✓ c26_FactoryEnergySystem.java	62.5 %	5	3	8
> EnergyMonitor	62.5 %	5	3	8
> c26_FactoryEnergySystem		0	0	0

5. Fuzz testing instruction coverage: 95.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	2.7 %	470	17,142	17,612
✓ src	2.7 %	470	17,142	17,612
✓ code	2.7 %	470	17,142	17,612
✓ c26_FactoryEnergySystem.java	95.5 %	470	22	492
> EnergyMonitor	77.1 %	64	19	83
> c26_FactoryEnergySystem	99.3 %	406	3	409

6. Z3 branch coverage: 62.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.6 %	5	835	840
✓ src	0.6 %	5	835	840
✓ code	0.6 %	5	835	840
✓ c26_FactoryEnergySystem.java	62.5 %	5	3	8
> EnergyMonitor	62.5 %	5	3	8
> c26_FactoryEnergySystem		0	0	0

7. Z3 instruction coverage: 95.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	2.1 %	470	21,630	22,100
✓ src	2.1 %	470	21,630	22,100
✓ code	2.1 %	470	21,630	22,100
✓ c26_FactoryEnergySystem.java	95.5 %	470	22	492
> EnergyMonitor	77.1 %	64	19	83
> c26_FactoryEnergySystem	99.3 %	406	3	409

8. Unimproved V-method branch coverage: 62.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.6 %	5	835	840
✓ src	0.6 %	5	835	840
✓ code	0.6 %	5	835	840
✓ c26_FactoryEnergySystem.java	62.5 %	5	3	8
> EnergyMonitor	62.5 %	5	3	8
> c26_FactoryEnergySystem		0	0	0

9. Unimproved V-method instruction coverage: 95.5%

Element		Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼  VibrationTestingTool		 2.1 %	470	21,630	22,100
▼  src		 2.1 %	470	21,630	22,100
▼  code		 2.1 %	470	21,630	22,100
▼  c26_FactoryEnergySystem.java		 95.5 %	470	22	492
>  EnergyMonitor		 77.1 %	64	19	83
>  c26_FactoryEnergySystem		 99.3 %	406	3	409